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COMMERCIAL ITEM DESCRIPTION

MANDRELS, MACHINE, SOLID AND EXPANSION

This Commercial item description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE

1.1 Scope. This description covers mandrels used for supporting workpieces between centers for machining purposes and inspection operations. This description is not intended to include all commercially available mandrels but only those commonly purchased by the Government.

1.2 Classification. This description covers mandrels of the following types and of the sizes listed in tables I thru III. The type and size to be furnished shall be as specified (see 6.2).

Type I - Solid mandrel: a cylindrically shaped tool with a 0.0005 inch per inch of length taper on the overall length, centered on each end, with a milled flat on one end for a holding device (see table I).

Type II - Expanding mandrel: a cylindrically shaped tool consisting of a solid arbor having a specific taper over the major portion of its length and one or more expanding sleeves with an inside taper which fits the tapered portion of the arbor. The arbor is centered on each end with a milled flat on one end for a holding device. When the sleeve is moved along the length of the arbor, it will contract or expand as determined by the location on the arbor. The plane of the outside surface of the sleeve shall be parallel with the horizontal center line of the arbor (see table II).

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Type III - Expanding mandrel, step type: essentially the same as the type II expanding mandrel described above. In addition, the step mandrel shall be manufactured with one or more steps in the sleeve (see table III).

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals, form a part of this description to the extent specified herein. The documents should be obtained from the contracting activity or as directed by the contracting officer.

Federal Specification PPP-T-1150	- Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery, Packaging and Packing of.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.
American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.

3. SALIENT CHARACTERISTICS

3.1 Design. The mandrels shall be new and of the manufacturers standard commercial design in effect on the date of the solicitation.

3.2 Material. The material used in the manufacture of the mandrels shall be steel suitable for the purpose intended.

3.3 Measurement system. All dimensions in this description are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.4 Identification marking. Each mandrel shall be permanently marked in accordance with the manufacturer's standard practice, including the size.

3.5 Hardness. The type I mandrels shall have a hardness of not less than 60 or more than 65 on the Rockwell "C" scale. The types II and III arbors shall have a hardness value of 51 to 62 and the sleeves shall have a value of 38 to 47 on the Rockwell "C" scale.

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TABLE I. Type I, solid mandrels.

Size	Diameter 1/ (inches)	Length 2/ (inches)	Size	Diameter (inches)	Length (inches)	Size	Diameter (inches)	Length (inches)
1	1/8	2 1/2	19	7/8	6 1/2	37	2	11
2	5/32	3	20	15/16	6 3/4	38	2 1/8	11 1/2
3	3/16	3	21	1	7	39	2 1/4	12
4	7/32	3 3/4	22	1 1/16	7 1/4	40	2 3/8	12
5	1/4	3 3/4	23	1 1/8	7 1/2	41	2 1/2	12 1/2
6	9/32	4	24	1 3/16	7 3/4	42	2 5/8	12 1/2
7	5/16	4	25	1 1/4	8	43	2 3/4	13
8	11/32	4 1/4	26	1 5/16	8 1/4	44	2 7/8	13
9	3/8	4 1/4	27	1 3/8	8 1/2	45	3	13
10	13/32	4 1/2	28	1 7/16	8 3/4	46	3 1/4	13 1/2
11	7/16	4 1/2	29	1 1/2	9	47	3 1/2	14
12	15/32	5	30	1 9/16	9 1/4	48	3 3/4	14 1/2
13	1/2	5	31	1 5/8	9 1/2	49	4	15
14	9/16	5 1/4	32	1 11/16	9 3/4			
15	5/8	5 1/2	33	1 3/4	10			
16	11/16	5 3/4	34	1 13/16	10 1/4			
17	3/4	6	35	1 7/8	10 1/2			
18	13/16	6 1/4	36	1 15/16	10 3/4			

1/ The diameter of the mandrel shall conform to the dimensions specified above when measured at a point located between L/3 and L/2, where L is the mandrel length, measured from the small end.

2/ Tolerances for length shall be +1/2 inches.

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TABLE II. Type II, expanding mandrels.

Size	Sleeve expansion range 1/ (inches)	Arbor length 2/ (inches)	Sleeve length 3/ (inches)
1	3/8 to 1/2	4 1/2	1 1/4
2	1/2 to 9/16	5	2 1/2
3	1/2 to 11/16	5 3/4	1 1/2
4	9/16 to 21/32	6	2 3/4
5	21/32 to 3/4	6	2 3/4
6	11/16 to 1	9 1/4	2 1/2
7	3/4 to 7/8	7	3 1/4
8	13/16 to 1	9 1/4	2 1/2
9	7/8 to 1	7	3 1/4
10	1 to 1 3/8	9	4
11	1 to 1 9/16	12 1/4	3 1/4
12	1 1/4 to 1 5/8	9	4
13	1 1/2 to 2	11 1/2	5
14	2 to 2 3/4	14	5 1/2
15	2 3/4 to 3 3/4	17	6
16	3 3/4 to 5	20	7
17	5 to 6 1/2	24	8

1/ Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

2/ Tolerance for arbor length shall be ±1/2 inches.

3/ Tolerance for sleeve length shall be ±1/4 inches.

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TABLE III. Type III, expanding mandrels, step type.

Size	Sleeve expansion range (inches) 1/	Arbor length (inches) 2/	Sleeve length 3/ (inches
1	1/2 to 1	9 1/4	3 1/4
2	1 to 1 1/2	10 1/2	4
3	1 1/2 to 2	12	4 3/4
4	2 to 3	17	7
5	3 to 4	17	7

1/ Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

2/ Tolerances for arbor length shall be ±1/2 inches.

3/ Tolerances for sleeve length shall be ±1/4 inches.

3.6 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers employees to undue hazards from the recovered materials.

3.7 Tolerances and finishes. Mandrel centers shall be machined and lapped to a standard included angle of 60 degrees ±15 minutes. The surface roughness of the mandrels shall not be greater than 16 microinches arithmetical average. The maximum allowable run out shall be not greater than 0.0005 inch total indicator reading for the type I and 0.001 inch total indicator reading at the grind diameter for types II and III. Sleeves of types II and III shall be finished internally to fit the arbor and shall provide uniform expansion to any size in the range of diameters listed in tables II and III.

3.8 Workmanship. Workmanship of the mandrels shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this description where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. The supplier shall randomly select the number of samples specified in table IV that is applicable to the quantity of item being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with the salient characteristics specified herein. Any sample failing to meet these characteristics shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table IV, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table IV, the entire lot shall be considered defective and unsuitable for shipment.

TABLE IV. Sampling plan.

Lot size	Sample size	Acceptance, number defective	Rejection, number defective
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6

4.3 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), preservation, packing, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

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